

Recovery Electronics Full Test

Title:	Electronics Full Test Report
Project:	HELVETIA
Filename:	Test_Report_Helvetia_Recovery_Electronics
Prepared by:	Fabian Kolodziej, Dongkyu Lee, Xuanting Meng
Checked by:	Alexandros Karadeloglou, François Martin-Monier

TEST DATA

General Information			
Test – ID:	ReEI01		
Subteam:	Recovery	Subsubteam:	Electronics

Introduction

This test report includes all functionality, quantity and partial system tests that have been conducted during the testing campaign. These also encompass partial system analyses with a focus on electronics for tests that already have a general test report. The report aims to present procedural aspects, test data and explain design choices.

Table of Contents

TEST DATA.....	1
Inhalt.....	Error! Bookmark not defined.
1 TEST Preparation.....	3
1.1 Attending People.....	3
1.2 Safety Concept	3
1.3 Materials.....	3
1.4 Location.....	4
1.5 Setup	6
2 MAIN TESTS	8
2.1 Functionality Tests	8
2.1.1 Auxiliary Board Functionality Test.....	8
2.1.2 PWM Modulation Test	11
2.1.3 Ground Tests	15
2.1.4 Lift-off Detection and Flight Mode Test	19
2.1.5 Battery Power Test	25
2.2 Reliability Tests	26
2.2.1 System Reliability Test.....	26
2.3 Partial System Tests and Analyses	29
2.3.1 Argos Launch	29
2.3.2 Helicopter Drop Tests (HDT).....	33
2.3.3 Ejection Tests	40
3 Continuation	41
3.1 Reusability of Parts and Material.....	41
3.2 How to Continue.....	41

1 TEST Preparation

1.1 Attending People

Name	Function
Fabian Kolodziej	Test Execution
Dongkyu Lee	Test Logging
Xuanting Meng	Test Execution

1.2 Safety Concept

1.2.1 Safety Gear

Gear	Further specifications	Status
Protective Glasses		☐
Protective Gloves		☐

1.3 Materials

1.3.1 Rocket Parts

Part/Assembly	Responsible Subteams	Status
Auxiliary Board	Recovery	☐
Flight Computers	Recovery	☐

Arming Electronics	Recovery	☐
Valves, Servo Motors	Recovery	☐
Power Switching Boards	Recovery, Avionics	☐

1.3.2 Equipment

Equipment	Further specifications	Status
Multimeter	A portable oscilloscope which doubles as a multimeter	☐
LiPo-Charger		☐
Oscilloscope		☐
Laptop	With required software and cables	☐
Camera		☐
Bench Power Supply		☐

1.4 Location

Short Description of the Test Location
The test locations were different for each test performed. Functionality and quantity tests were all conducted in free working spaces that provide basic electronics testing equipment.

Provided at the Test Location

Testing equipment which includes the equipment stated in 1.3.3.

Possible Environmental influx to the Test at the Location and vice versa

No general influx on functionality.

Signal interference by devices and metallic structures in the vicinity.

1.4.1 Environmental Conditions

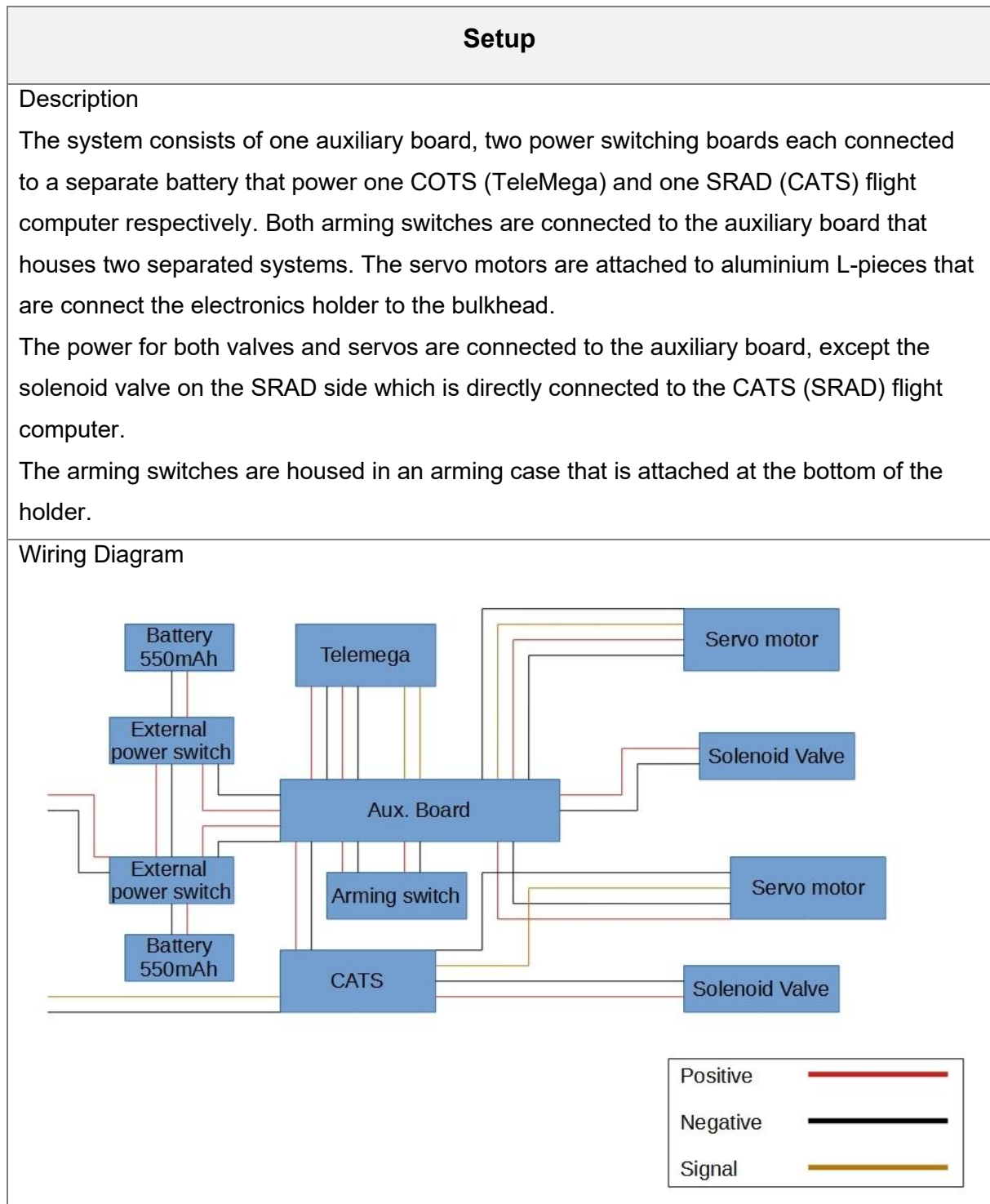
	Actual values	Acceptable range
Weather	Inside	Any
Temperature	20°C	Comfortable for People
Other notes		

1.4.2 Comparison to the Launch Condition

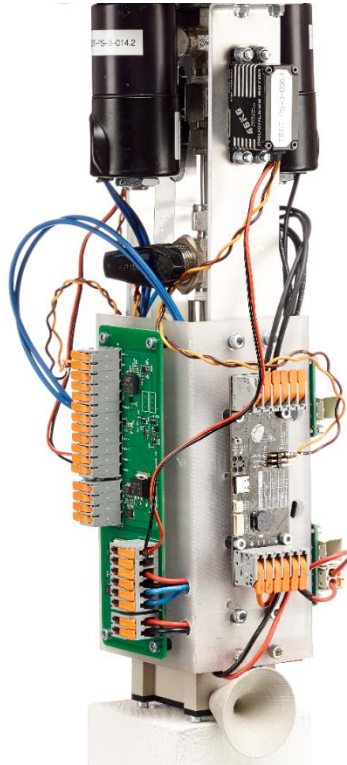
Different Location -> Different GPS Signals

Sand could get into connectors so that assembly could prove to be more difficult

1.5 Setup



Picture



1.5.1 Pre-Test Checks for the Setup

What needs to be checked to ensure the success of the test?

Item	Comments	Status
Assembly Procedure		☐
Arming Checklist		☐

2 MAIN TESTS

2.1 Functionality Tests

Setup:

For general functionality tests of our system, the relevant parts will be assembled as necessary. The basic setup is described and explained in the previous chapter. Each unique setup is illustrated with a picture at the beginning of each individual test subchapter.

Purpose of Test:

The following tests are performed to test and confirm the capability and functionality of the system. This is to identify any flaws in the current concept and design. A functionality test of the entire system requires a variety of different tests that validate small parts of the system so that problems can only occur in isolated and controlled instances.

2.1.1 Auxiliary Board Functionality Test

Purpose of Test:

During the armed state, all in- and outputs of the board are strictly defined and need to be within an expected range. As such, probing the board for its voltage levels and potential bridges is imperative to identify any electrical errors that could later lead to greater failure modes and cause harm to connected devices.

Functionality Checklist:

Procedure	Value	Status
Battery Connector Inputs '+' and '-' must not be bridged	Twice for both circuits	☐
After connecting batteries to input, voltage between input pins needs to be measured	12V	☐
Test arming circuits by shorting 'Arm+' and 'Arm-' with a jumper cable	12V	☐
Measure voltage between 'CATS+' and 'CATS-'	12V	☐
Measure voltage between 'Motor+' and 'Motor-'	12V	☐
Measure voltage between 'Pyro+' and 'Pyro-'	12V	☐
Measure voltage between both 'Valve' connectors	0V	☐
Measure voltage between 'Tele+' and 'TM-'	3.7V	☐

Results:

Parameters	Value
U(+, -)	12.2V (LiPo)
Arming Circuit	Non-functional
U(CATS+, CATS-)	9.78V
U(Motor+, Motor-)	9.78V
U(Pyro Battery, GND)	9.78V

U(12V, GND)	9.78V
U(3.7V, GND)	3.716V
LED's	All Functional

Conclusion:

The arming did not work as intended when opening and shorting the arming inputs. The MosFET's did not open when the arming switches were turned off. This lead to the assumption that either the MosFET's were malfunctional or the circuit design had an issue. It turns out that the manufacturer PCB design files were wrong. The Drain and Source were interchanged for both schematics and footprints so that the MosFET effectively let current through its protector diode.

Mitigation:

After identifying the issue, the amount of LED's has been increased for safety and easier problem detection. Furthermore, the schematics and footprints have been corrected in a new version of the board (RecoveryBoard_V3).

Results of Retest of V2 boards:

Parameters	Value
U(+, -)	12.2V (LiPo)
Arming Circuit	Functional
U(CATS+, CATS-)	12V

U(Motor+, Motor-)	12V
U(Pyro Battery, GND)	12V
U(12V, GND)	12V
U(3.7V, GND)	3.716V
LED's	All Functional

Conclusion:

With the new version of the boards, the arming circuit was functional and the MosFET switched with no issues. Now instead of the questionable 9.7V, we got the full 12V provided. As such, the full general power functionality of the board was confirmed.

2.1.2 PWM Modulation Test

Purpose of Test:

For the deployment of the main parachute, the second deployment event, a pulse-width modulation (PWM) signal is used. This in turn controls the position of the servo motor arm. As the PWM outputs on the TeleMega (COTS) flight computer have not been fully implemented yet, a PWM modulator circuit has been implemented on the auxiliary board and needs to be tested. Due to the servo motors missing datasheets, the PWM range is yet to be identified. For this reason, trimmers have been implemented on the board to match the PWM signal to any range between 0% to 100%.

Setup change:

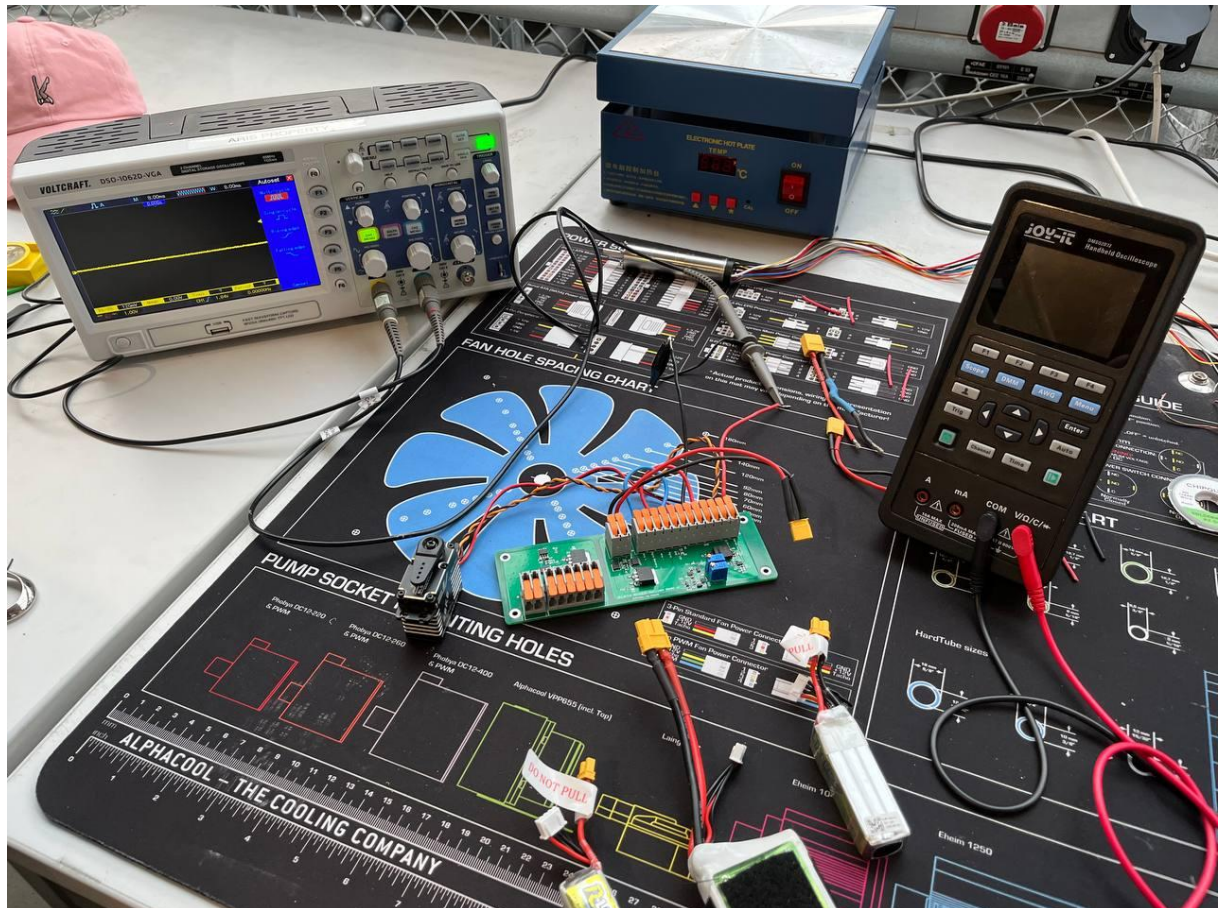


Figure 2.1 Servo Motor PWM Input connected to an oscilloscope

PWM Modulation Test Procedure:

Procedure	Value	Status
Pull the Arming Pin	System Armed	☐
Connect an oscilloscope to the connectors 'Servo' and 'GND'		☐
Measure the signal for position 1 (Default position)	Servo Default	☐

Turn the trimmer to see changes and try to match signal to 5% duty (usual minimum value of conventional PWM signal ranges)		☐
Connect connector 'Pyro' to 'GND'		☐
Measure the signal for position 2 (Deployment active)	Servo Deployed	☐
Turn the trimmer to see changes and try to match signal to 10% duty (usual maximum value of conventional PWM signal ranges)		☐
Insert Arming Pin	System Disarmed	☐

Results:

Parameters	Value
Frequency	50.12Hz
Voltage Logic High	3.7V
Voltage Logic Low	0V
PWM Duty Cycle 1	15%
PWM Duty Cycle 2	25%
Servo Arm Position 1	Out of Range
Servo Arm Position 2	Out of Range
MosFET 1	Non-Functional
MosFET 2	Functional

Conclusion:

As the switching between the 'Default' and 'Deployed' modes was not working as intended, the switching power MosFET's were investigated. It turned out that the MosFET 1 being non-functional was due to the gate driver. The output of the gate driver was lower than what it stated in the datasheet, which led to the MosFET 1 switch never opening and as a result, never switching.

Therefore, for 'Deployed' mode, when 'Pyro' is grounded, the resistors were set in parallel and thus duty increased instead of getting controlled by a set trimmer.

Furthermore, even during 'Default' mode in which MosFET 2 is open and MosFET 1 is closed, the duty was out of the operating range of the servos.

The cause is the resistor at the 'MOD' input of the LTC6992 IC being too large which led to a higher voltage which in turn leads to higher duties.

Mitigation:

By removing the 10k resistor at the input of the gate driver, both voltages on input and output sides of the driver are increased. Consequently, the voltage difference between Gate and Source lies below the threshold voltage of 0.8V and the MosFET switches.

As the voltage across the resistor at the MOD input of the LTC6992 IC and the duty are closely related, changing the 1.2k resistor to a 330Ohm should also lower the resulting duty and thus the duties should enter the operating range of the servo motors.

Results of Retest (V2):

Parameters	Value
Frequency	50.12Hz

Voltage Logic High	3.7V
Voltage Logic Low	0V
PWM Duty Cycle 1	2%
PWM Duty Cycle 2	13%
Servo Arm Orientation 1	Up
Servo Arm Orientation 2	Down
MosFET 1	Functional
MosFET 2	Functional

Conclusion:

With the new changes, the PWM modulating circuit on the auxiliary board V2, successfully switched and controlled the servo motors so that we got two stable positions for the servo motor arms.

2.1.3 Ground Tests

Purpose of Test:

To ensure safe and reliable deployment of both drogue and main parachutes, the system needs to be sufficiently tested on the ground before it may be used in any real flight test. The capability of the system to carry out the mission in its full capacity is essential and most of it can be tested on the ground.

The CATS (SRAD) has been tested by setting up a timer to trigger the events by logic. As for the TeleMega (COTS), all ground tests can be done by manually triggering the pyro channels in question.

Setup Change:

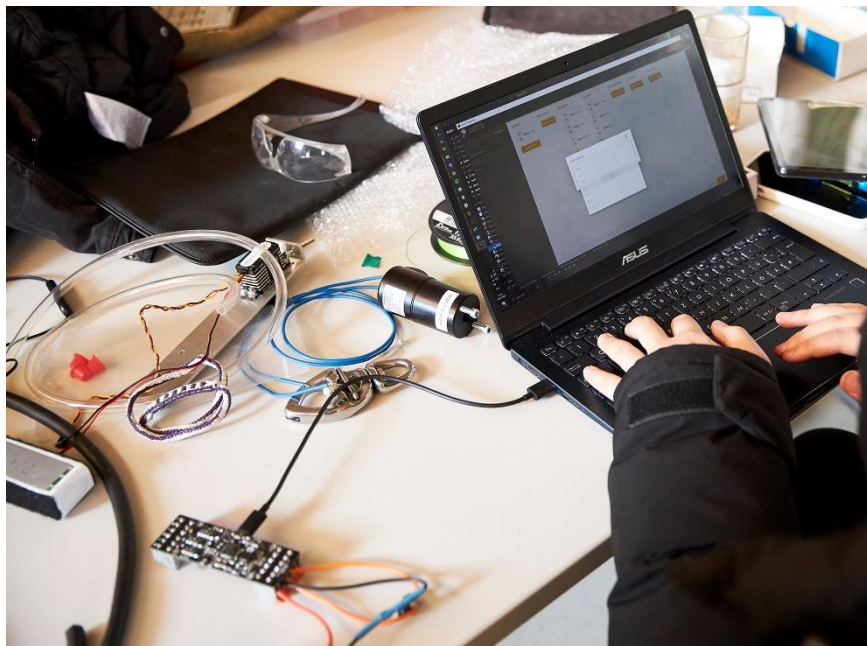


Figure 2.2 Programming CATS (SRAD) FC for ground tests. The valves and Motors can be directly connected to flight computer.

Ground Test Procedure (CATS):

Procedure	Comments	Status
Configure CATS (SRAD) Flight Computer (Ground Test configuration)		☐
Pull the Arming Pin	System Armed	☐
Wait for Ready State of the CATS Flight Computer		☐
Wait 10 seconds for Solenoid Valve Actuation	Drogue Deployed	☐

Wait 5 more seconds for Servo Motor Actuation	Main Deployed	☐
Insert Arming Pin	System Disarmed	☐

Results:

Parameters	Value
Valve Actuation	Successful (3s)
Servo Motor Actuation	30° to 150°

Conclusion:

Valve actuation worked flawlessly for 3 seconds as configured. The servo motor actuation does not cover the entire range of the servo. This however was expected, as the duty range of the CATS PWM output is from 5% to 10% whereas the servos seem to operate from 2% to 13%. This is the estimated range from the results obtained in the PWM Modulation Test.

With the system connected to the servos and the snap shackle, the range of 30° to 150° was still enough to open the snap shackle. Furthermore, by shortening the lines between the snap shackle and the servos, this difference in frequency range can be overcome.

Ground Test Procedure (TeleMega):

Procedure	Comments	Status
Connect TeleMega (COTS) to a notebook computer via Micro USB cable and orient the board horizontally.		☐

Pull the Arming Pin	System Armed	☐
Confirm Idle State and Battery Percentage reported by beeps.		☐
Select Manual Pyro Trigger on the TeleMega Software Interface		☐
Fire Pyro Channel D to actuate Solenoid Valve	Drogue	☐
Fire Pyro Channel B to actuate Servo Motor	Main	☐
Insert Arming Pin	System Disarmed	☐

Result:

Parameters	Value
Valve Actuation	Successful (3s)
Servo Motor Actuation	0° to 180°

Conclusion:

For the TeleMega, ground tests were performed by manual trigger which in this case worked flawlessly. Due to the nature of manual triggers, the events could only be triggered sequentially. However, that corresponds to the needs of the mission and the TeleMega cannot in either way trigger two events simultaneously. Therefore, the technical capabilities and functionality of the TeleMega combined with the auxiliary board could be validated.

2.1.4 Lift-off Detection and Flight Mode Test

Purpose of Test:

Whilst the ground tests provide the validation of the technical capabilities of the design and the system components, the telemetry and logic components have not been tested in combination with the mentioned system. Hence, it makes sense to simulate a flight environment as much as possible and test the event-triggering.

Setup Change:

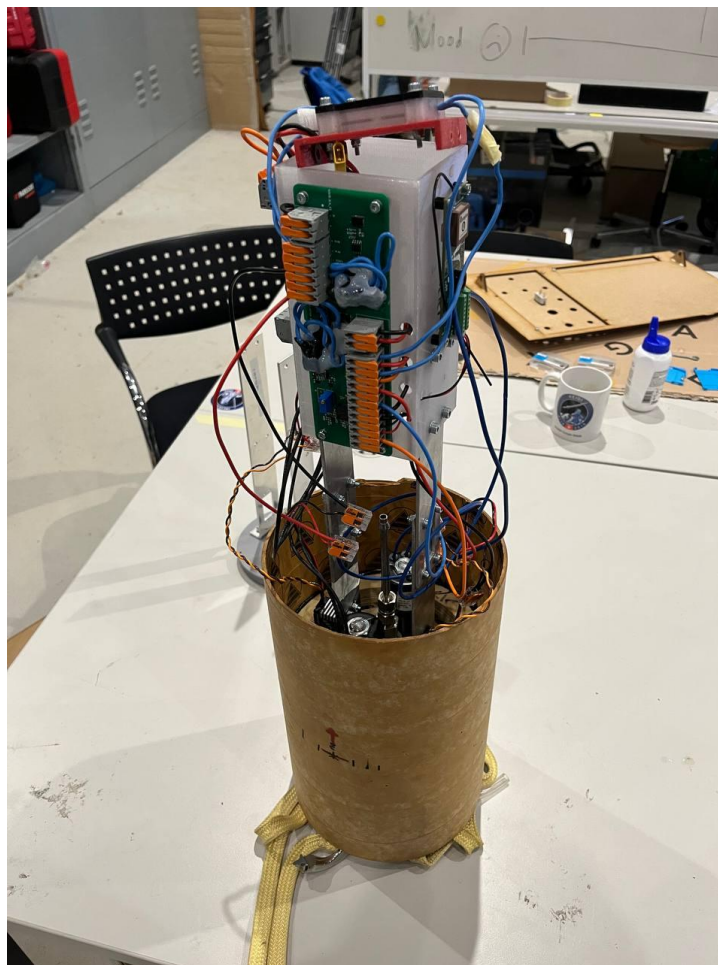


Figure 2.3 Setup of Argos Launch. Most convenient and structurally rigid setup for simulated flight mode tests.

Lift-off and Flight Mode Test Procedure (CATS):

Procedure	Comments	Status
Configure CATS (SRAD) Flight Computer (Flight Test configuration)		☐
Pull the Arming Pin	System Armed	☐
Confirm 'Ready' state of CATS	Ready	☐
Pull the system up with force		☐
Confirm Valve Actuation	3 seconds	☐
Confirm Servo Actuation	Down	☐
Insert the Arming Pin	System Disarmed	☐

Results:

Parameters	Value
Lift-off Detection	Successful
Valve Actuation	3 seconds
Servo Actuation	30° to 150°
Arming	Successful

Conclusion:

As the CATS (SRAD) flight computer does not provide direct methods for manually triggering deployment events, the flight mode did not differ much from the afore performed ground tests. This test was mostly to see whether

the Lift-off detection worked in combination with the telemetry and logic of the CATS flight computer.

Lift-off and Flight Mode Test Procedure (TeleMega):

Procedure	Comments	Status
Configure TeleMega (COTS) Flight Computer (Flight Test configuration)		☐
Pull the Arming Pin	System Armed	☐
Confirm 'Flight Ready' state of TeleMega	Flight Ready	☐
Pull the system up with force		☐
Confirm Valve Actuation	3 seconds	☐
Confirm Servo Actuation	Down	☐
Insert the Arming Pin	System Disarmed	☐

Results:

Parameters	Value
Lift-off Detection	Successful
Valve Actuation	Malfunctional
Servo Actuation	Non-Nominal, Shaky Servo Arm
Arming	Successful

Observation:

Both deployments did not operate nominally. For both the valve and the servo motor actuation, the pyro channel signals on the TeleMega worked in a “PWM”-like state with high frequency. Such a behaviour would prevent the valves from releasing enough CO₂. In addition to that, the release of the snap shackle cannot be guaranteed due to the shaky nature of the servo arms turning. Thus, the test is considered a failure and the source of the problem needs to be identified as soon as possible. In this state, redundancy of the system is not given.

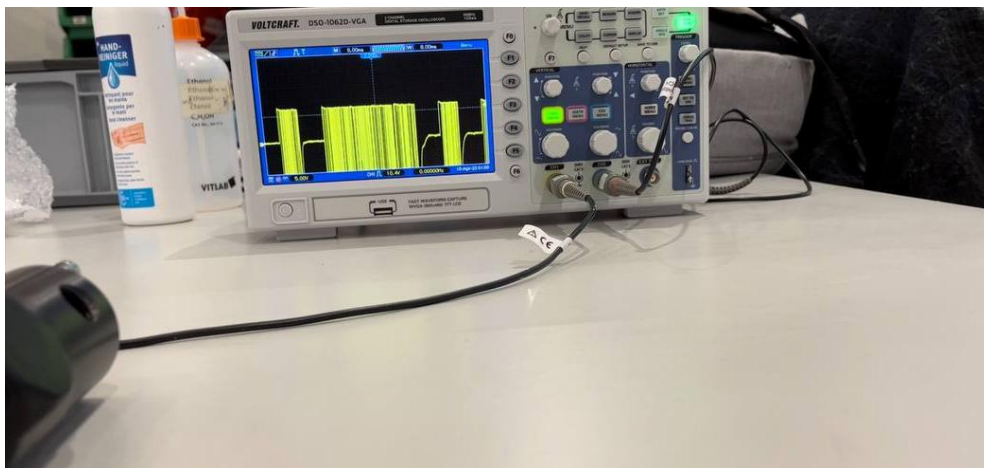


Figure 2.4

Furthermore, during some of the tests, a MosFET of one of the pyro channels blew up and shorted the pyro circuit (Pyro channel C on the TeleMega 4.0 and the 5.0 in particular). This alone does not impact the mission, in case of such behaviour during a rocket launch, as this only happens after valve actuation. As such, after valve actuation, the valve would simply not close anymore which means nosecone ejection is successful. However, that pyro channel would not be operational again which leads to only 3 out of 4 pyro channels

being operational on each TeleMega unit at the time of writing this report. At least 2 are necessary for a successful mission.

Mitigation:

The circuit lacked a flyback diode for protection in case inductive loads damaging the MosFET's were the reason for the shorting. Though the ground tests worked without problems, it seems the high frequency switching caused some cumulative damage to the MosFET's. This however has not happened to the pyro channels connected to the servo motors implying that it is the missing flyback protection causing the damage.

As for the high frequency switching itself, different batteries seem to cause different behaviour when observing the servo motor arms turning.

Battery Voltage	Turn Before Malfunctioning
11.08V	90°
12.59	150°

The higher battery voltage led to a longer interval before the high frequency switching started. Thus, it was inferred that there was a correlation between the malfunctioning MosFET's and the input voltage.

Apart from the internal gate signal which is regulated to a stable 3.28V by an LDO, the gate of the pyro channel MosFET's is also controlled by a differential dual comparator. It compares and pulls the gate down to ground for input power supplies at for less than 3.5V. As such, the pyros are locked and cannot fire at a voltage smaller than 3.6V, presumably due to voltage drop-offs and real-world circumstances. The TeleMega manual states that a Li-Po battery must provide a voltage of equal to or higher than 3.8V for the ground station to

give a 'GO' status for launch. However, telemetry, logic and even the pyros seem to fire at a power supply of the 3.716V the auxiliary board provided. After testing for different power supply levels, it could be identified that the high frequency switching only happened at 3.7V.

Battery Voltage	Test Result After 3 Consecutive Tests
3.6V	No Actuation
3.7V	High Frequency Switching Pyros
3.8V	Perfect Actuation

At 3.6V, the pyros did not fire at all, whereas at 3.8V, the pyros fired perfectly for both valves and servos. After discussing this matter with the creators of the TeleMega flight computer, this high frequency switching neither reflects nominal behaviour nor any expected failure modes of the TeleMega. They also stated that whilst the telemetry remains powered at power supply levels lower than 3.8V, the pyros are supposed to be locked per design.

Conclusion:

After changing a resistor on the auxiliary board to increase the TeleMega power supply from 3.716V to 4.4V, the pyros fired without problems. As the high frequency switching did not occur in 10 consecutive attempts, the possibility of it being linked to the auxiliary board has been ruled out. However, as this seemed to be a phenomenon that only ever occurred at 3.7V, it could be considered as mitigated by increasing the power supply voltage for the TeleMega. It may be possible that the high frequency switching is a result of the TeleMega reacting to a critical voltage and locking pyro channels.

2.1.5 Battery Power Test

Purpose of Test:

The system needs to stay powered through the entire pre-launch process which includes every step after arming the Recovery system: pre-filling, remote filling, decoupling and ignition. Prior to decoupling, the system gets powered by the external power provided by a power connection to the Avionics system. After decoupling, the power draw switches from the external power to battery power. From this point on, the system will be powered anywhere from a few minutes up to an hour until the signal for launch has been given. In order to ensure sufficient runtime of the system, the batteries need to be tested on how long they can last before dropping in voltage significantly.

Battery Power Test Procedure:

Procedure	Comments	Status
Confirm the Arming Pin is inserted		☐
Connect both batteries to the connectors	550mAh (2x)	☐
Pull the Arming Pin	System Armed	☐
Confirm active state of the system and wait whilst continuously probing the connector voltage to prevent over discharging	$U(\text{batt}) > 3.3\text{V}$	☐
After waiting at least 2 hours, perform ground tests to confirm functionality of system		☐
Insert the Arming Pin	System Disarmed	☐

Results:

Parameters	Value
Test Duration	3h30
Battery Voltage	3.786V; 3.735V
Valve Actuation Count	10
Servo Actuation Count	10

Conclusion:

The system was active for 3h30 and could still reliably provide enough current to actuate the valves reliably which draw 5A for 3 seconds. The battery voltage for both systems was between 3.7V and 3.8V. Thus, the batteries are large enough to power the system at least thrice the duration necessary.

2.2 Reliability Tests

2.2.1 System Reliability Test

Purpose of Test:

For the Recovery system, reliability is imperative for a successful mission. Therefore, reliability tests are of utmost importance to validate the entire system. The tests are performed by simulating lift-off in flight mode to imitate a real-flight environment.

The tests included in these reliability tests will follow the procedure of the lift-off detection and flight mode tests. Due to the redundant nature of the system, both parts need to be tested separately. The successes that have been

documented are the amounts of consecutive successes without failure since a single failure suggests a problem in the system. As such after each test failure, the problem has been successfully identified and mitigated. The data is therefore from the latest iteration of the system.

Test Data:

System Component	Consecutive Successes
Arming System	80
CATS (SRAD) System Functionality	50
TeleMega (COTS) System Functionality	50
Separation Mechanisms	150

Conclusion:

At 50 consecutive successes, we consider a system as reliable enough to be considered reliable. Both arming system and the separation mechanisms have been tested separately and together with the functionality tests which might explain the high numbers. These have never failed to work. As such, it stands to reason to consider these parts of the system as reliable as well.

2.2.2 Drone Test

Purpose of Test:

In order to have a system verification of any kind on the TeleMega branch of the redundant setup, drone tests were conducted to simulate a rocket take-off.

The flight will follow a simple ascent and descent motion to kickstart the state machine and flight data recording with lift-off.

In these tests, a servo motor arm was turned by detecting apogee and sending the drogue deployment signal which was difficult to observe with simple ground tests. The drone flight patterns were more in line with actual flight behaviour.

Setup Change:

The TeleMega branch is the only system to be tested. Therefore, the auxiliary board, together with the TeleMega FC and a servo motor to test and visualize the functionality were the only relevant systems. The boards were mounted on a CLA holder which was taped to the drone stand.

Drone Test Procedure:

Procedure	Comments	Status
Configure TeleMega (COTS) Flight Computer (Drone Test configuration)		☐
Confirm Safety Range in both airspace and on the ground		☐
Verify Stability of Test Setup		☐
Attach Setup to Drone with Servo Motor in camera's frame	A lot of Black Tape	☐
Start Drone and ascend slowly to about 50cm above ground		☐
Simulate Lift-off by abruptly ascending with max speed		☐
Simulate Apogee by stopping and descending at max speed		☐

Simulate Touchdown by Stopping descent

☐

2.3 Partial System Tests and Analyses

Note: General details on the following tests can be found in each respective test report.

This part of the report aims to analyse and discuss the test data.

2.3.1 Argos Launch

Purpose of Test:

Real rocket launch to test recovery system in flight.

Test Data and Analysis:

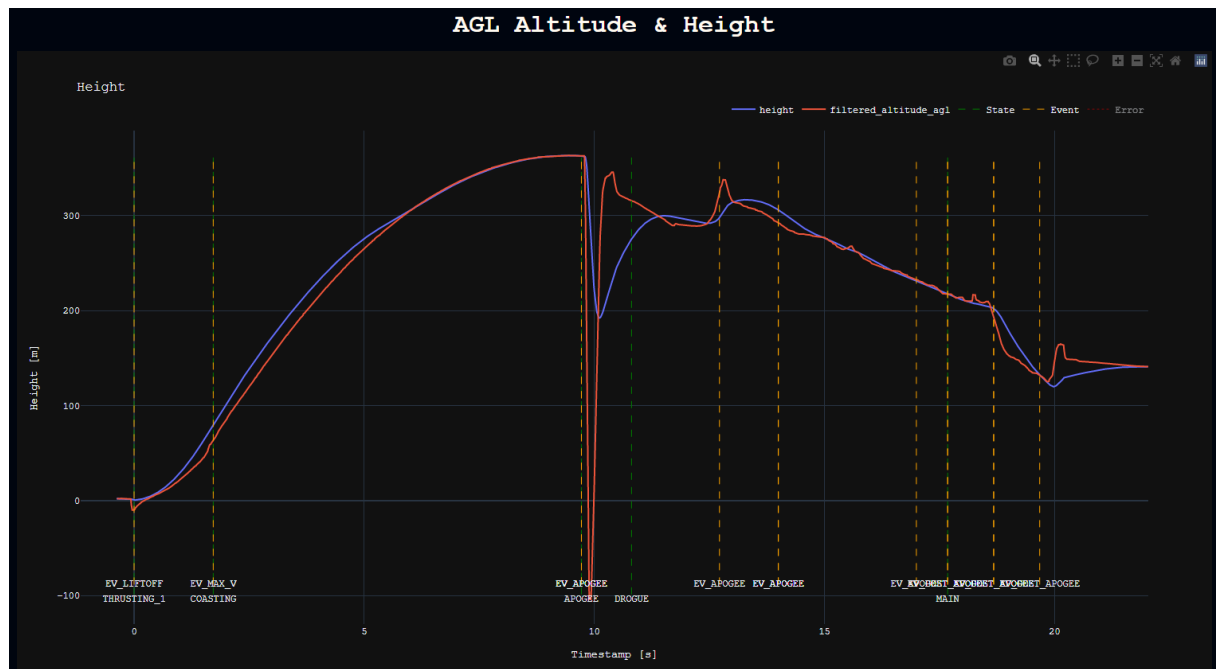


Figure 2.5 Argos: Altitude Data obtained from CATS (SRAD) FC

Figure 2.5 shows an apogee reached at around 1180ft (360m) even though the target altitude was at 1640ft (500m). This however was either due to inaccurate simulations or other real-world circumstances. Apogee was successfully detected, and the events triggered according to the configuration. One non-conformity is the sudden drop in altitude after the first deployment event for 0.5 seconds. This might be due to the pressure spike depicted in figure 2.6.

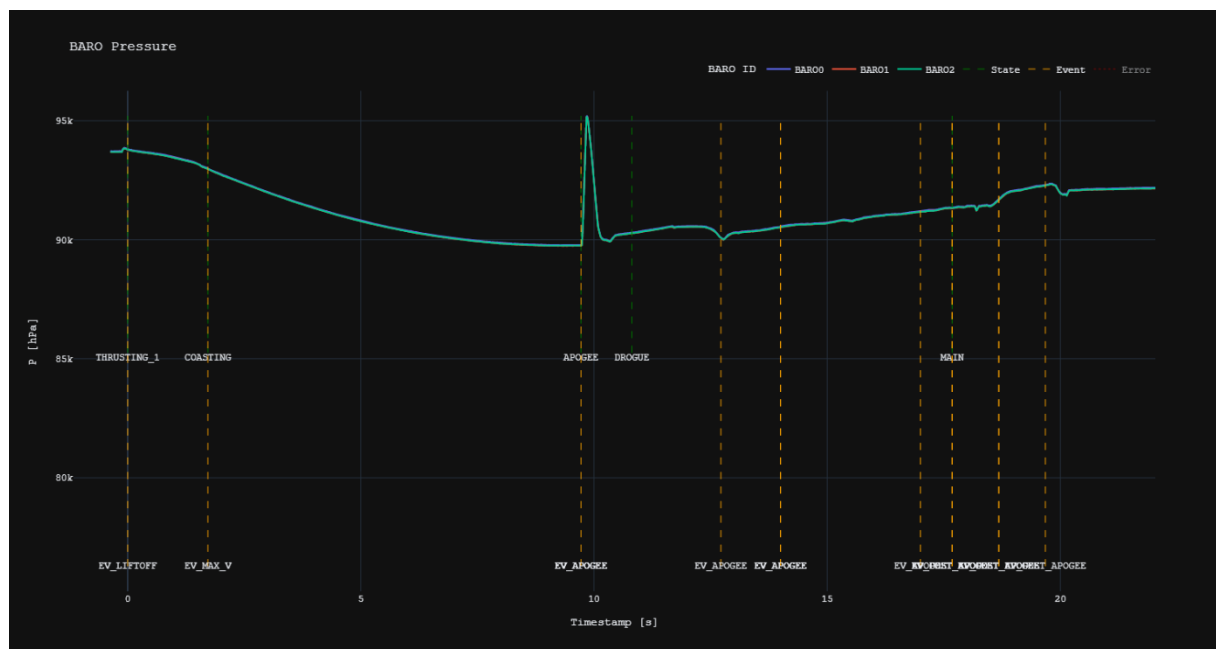


Figure 2.6 Argos: Pressure Data obtained from CATS (SRAD) FC

Main deployment was set to 720ft (220m) and would have been triggered. However, the CATS (SRAD) flight computer has a system implemented in its FSM which prevents the main parachute to be triggered by altitude alone. This is the reason why main state got detected at the real 720ft (220m).

shows how the igniter circuits have a constant voltage level, implying they never fired during flight.

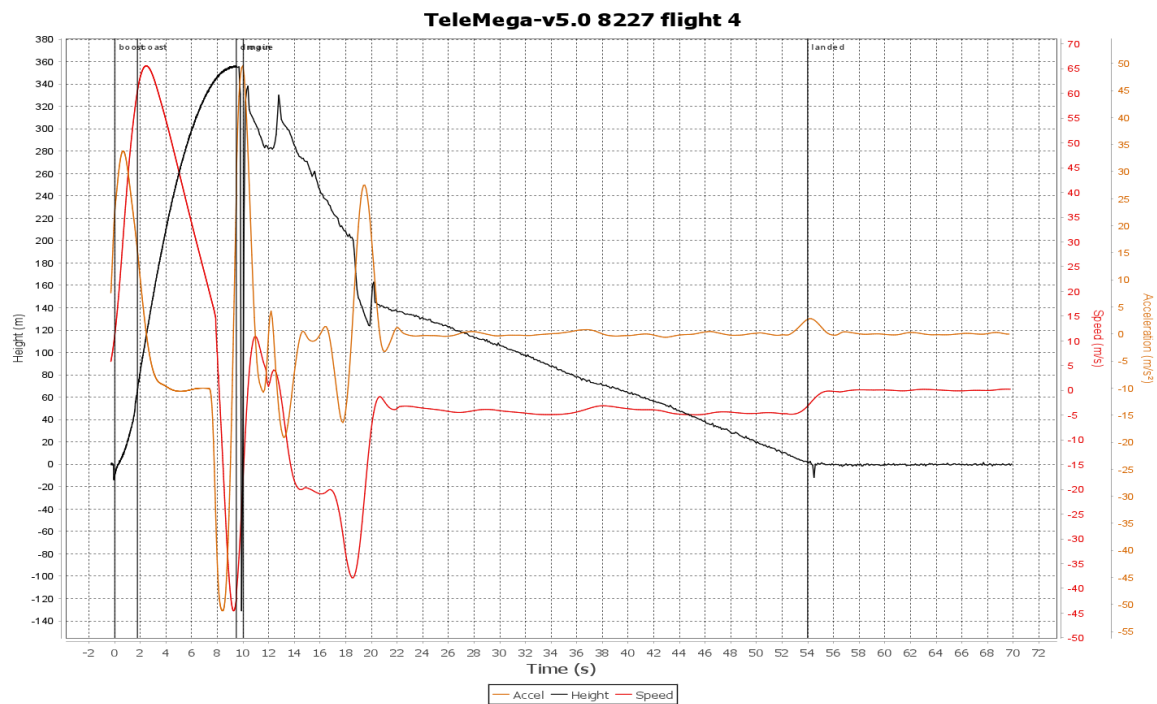


Figure 2.8 Argos: Height, Acceleration and Speed Data obtained from TeleMega (COTS)

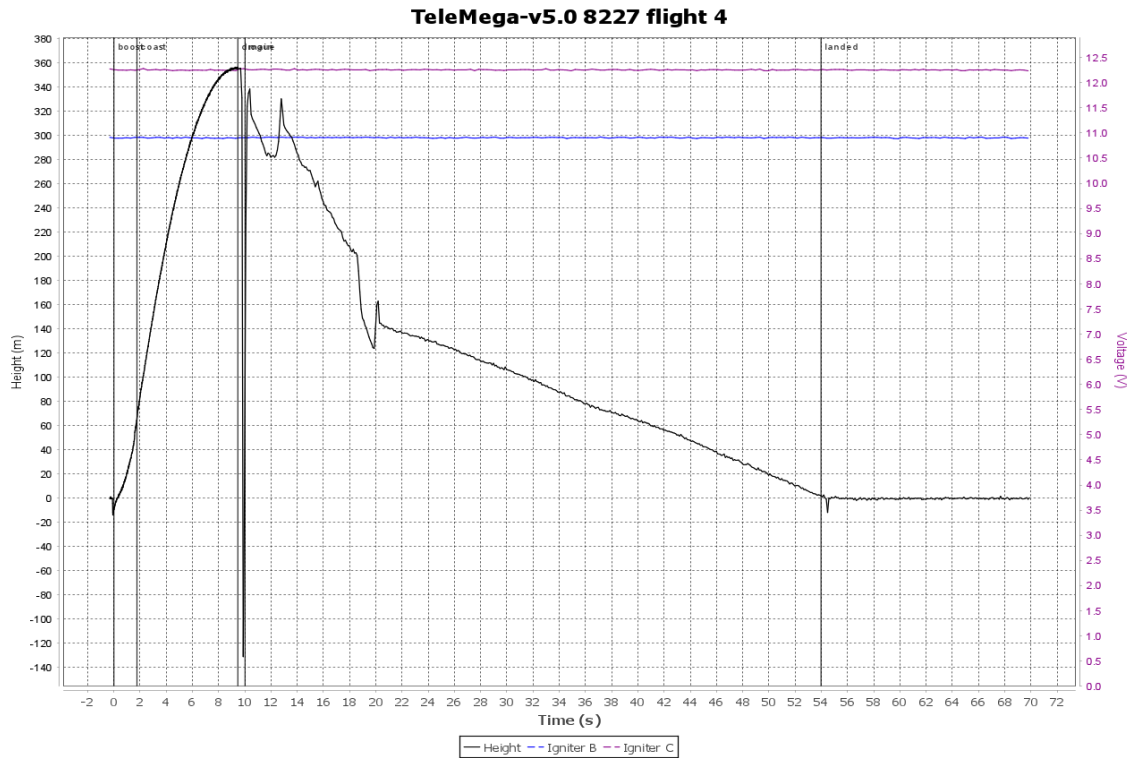


Figure 2.9 Argos: Valve (C) and Servo (B) Pyro Voltages

2.3.2 Helicopter Drop Tests (HDT)

Purpose of HDT 1:

Record flight data and monitor flight via COTS ground station hardware. GPS tracking was also tested.

Test Data and Analysis:

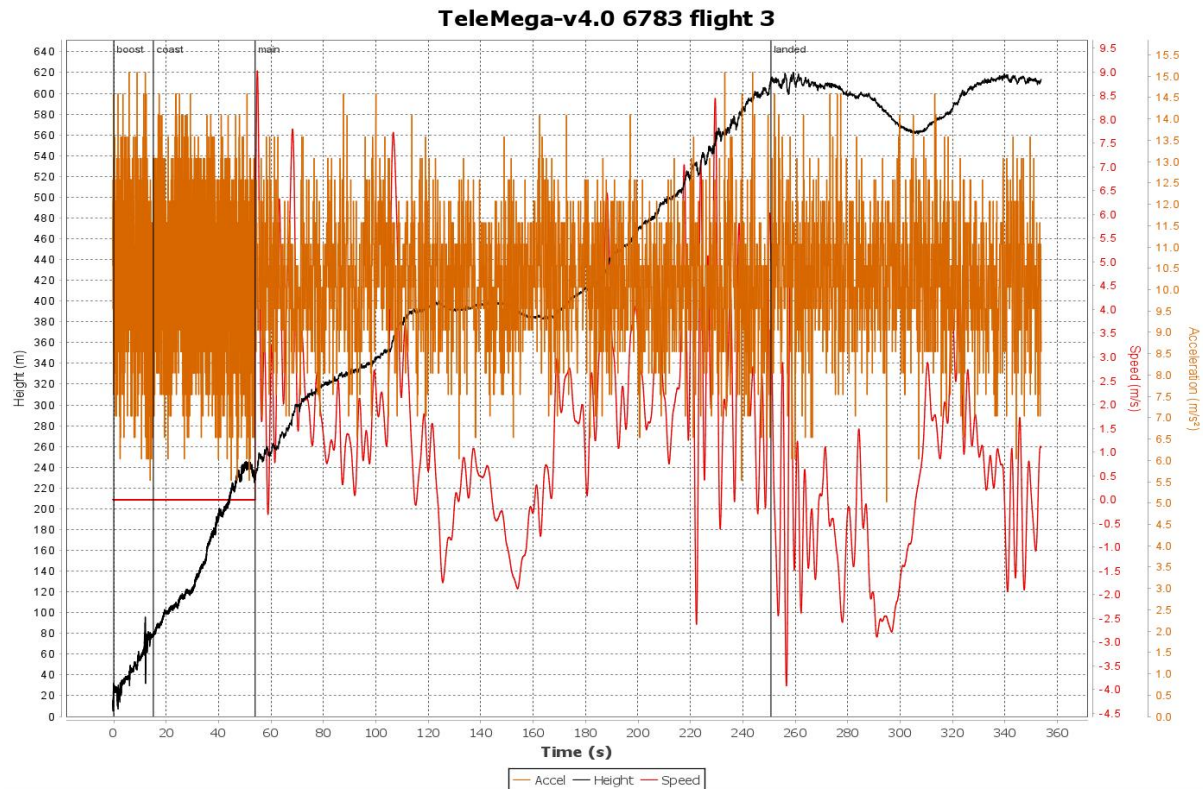


Figure 2.10 HDT 1: Height, Acceleration and Speed Data obtained from TeleMega (COTS) FC

The first HDT revolved around a completely mechanical system. In addition to that, the TeleMega was the only flight computer on board which was there for flight data logging and GPS tracking. Unfortunately, the helicopter had to stand still for a few seconds to wait for unauthorised personnel to leave the drop zone. As a consequence, the flight computer assumed 'landed' state and stopped logging data from that point on. Therefore, both GPS tracking and flight data stop after 355 seconds which can be observed in figure 2.10 and 2.11.

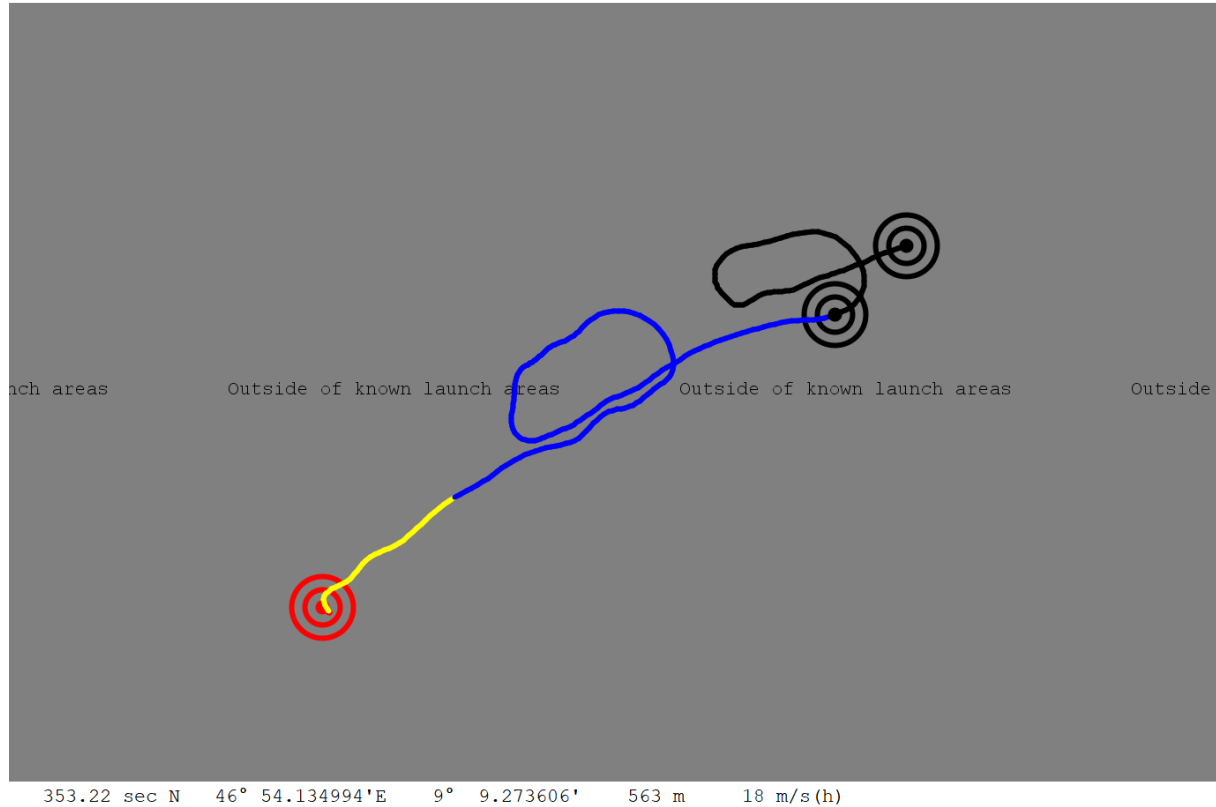


Figure 2.11 HDT 1: GPS Data obtained from TeleMega (COTS) FC

Purpose of HDT 2:

Remote triggering of 2nd deployment event to test main parachute shockload. Both SRAD and COTS flight computers log data. COTS flight monitoring and GPS tracking tested again.

Test Data and Analysis:

In the first 150 seconds, the graph follows the hovering of the helicopter until the drop happens at roughly 152 seconds. The lift-off and the beginning of the flight are not recorded, as the recorder got initiated by a timer from lift-off. The beginning of the recording corresponds to 200 seconds after lift-off got detected. The drogue and main descent are clearly depicted in the flight graph (figure 2.12).

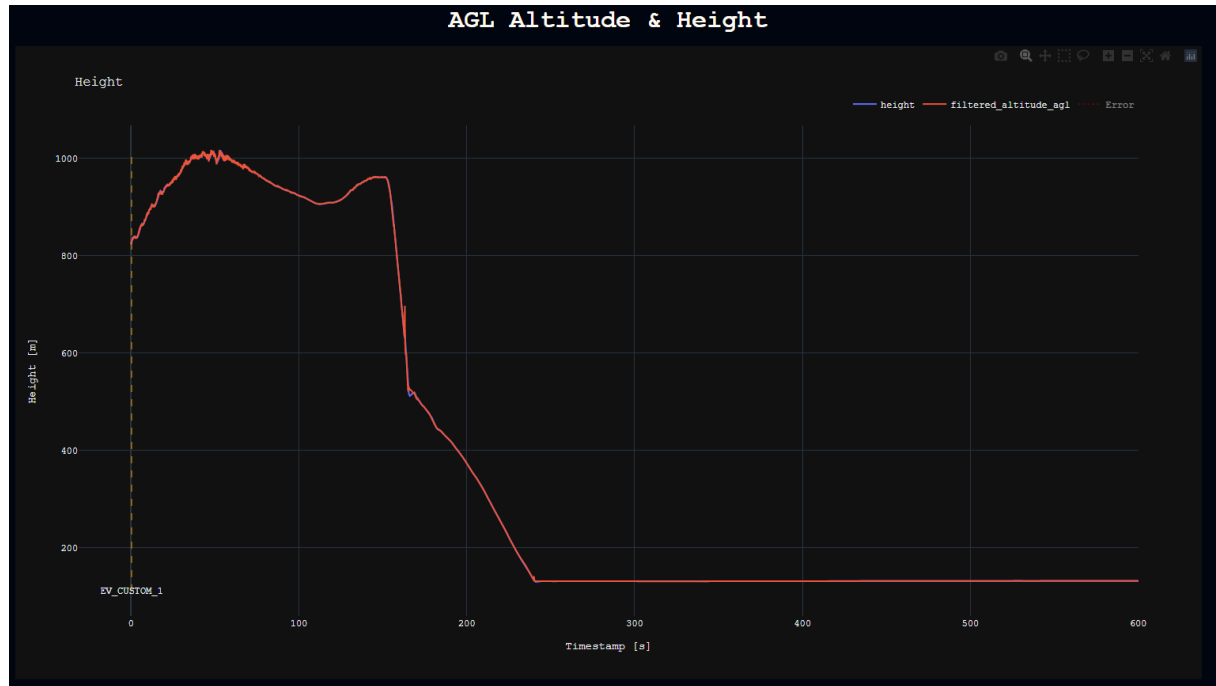


Figure 2.12 HDT 2: Altitude Data obtained from CATS (SRAD) FC

The drogue descent reached a maximum velocity of around 130ft/s (40m/s) at which it stabilised, and the main chute deployment got initiated remotely.

Figure 2.13 shows such velocity profile and indicates the main deployment as the sudden velocity spike up to 180ft/s (55m/s). This is the peak velocity the structure reaches during which the main chute falls out and the lines need to be taut. When main chute is fully opened at 165 seconds, the velocity drops to roughly 20ft/s (6m/s). From 170 seconds onwards it keeps this velocity until touchdown happens at 245 seconds. In figure 2.14, the shock load for the main chute deployment can be assessed as 12.9g which reflects the simulation data at 13g very well.



Figure 2.13 HDT 2: Velocity Data obtained from CATS (SRAD) FC



Figure 2.14 HDT 2: Acceleration Data obtained from CATS (SRAD) FC

The TeleMega (COTS) data seems to correspond very accurately to the CATS (SRAD) data. Figure 2.15 shows height, acceleration and speed in one graph. Like the CATS flight computer, the velocity goes up to a maximum of 127ft/s (39m/s) during drogue descent and the main chute shock load at deployment reaches 12.5g. Therefore, as both flight computers present almost identical data, the drop test proves that the entire recovery section is functional within expected and simulated range.

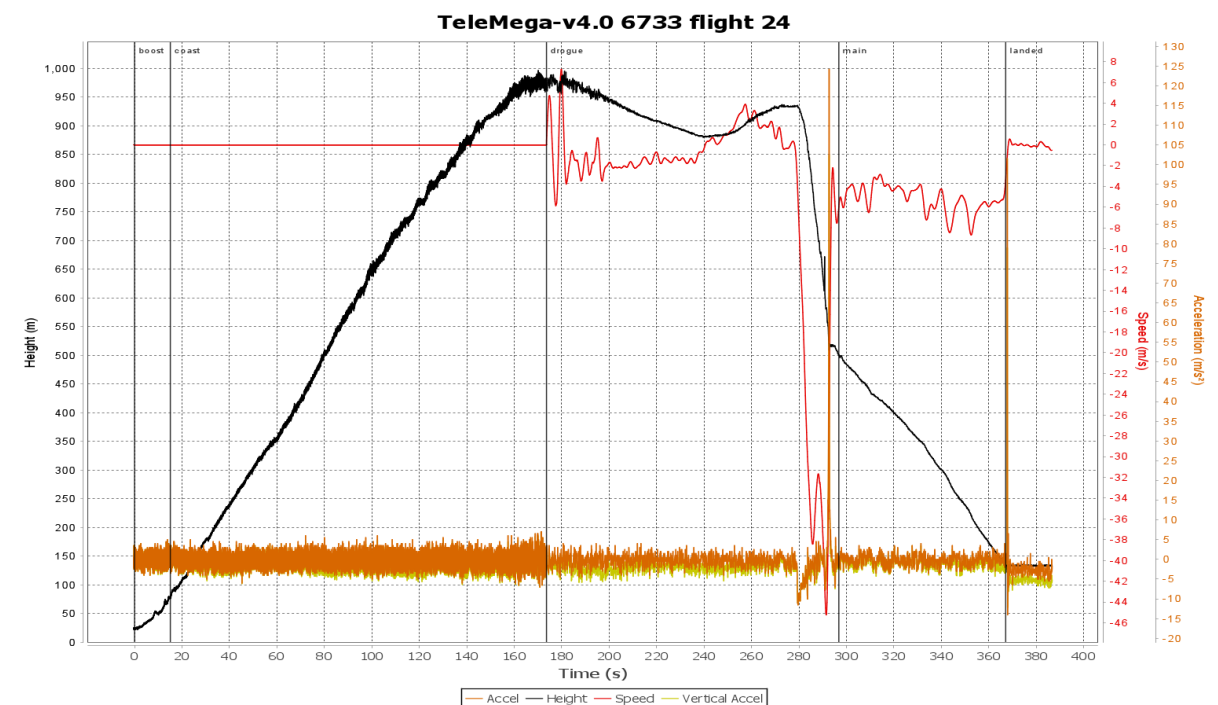


Figure 2.15 HDT 2: Height, Acceleration and Speed Data from TeleMega (COTS) FC

On top of that, the helicopter did not need to wait this time. Therefore, the TeleMega recorded until touchdown and GPS tracking could be confirmed until the very end with the accurate location. As a result, ground recovery was not an issue and could be found easily.

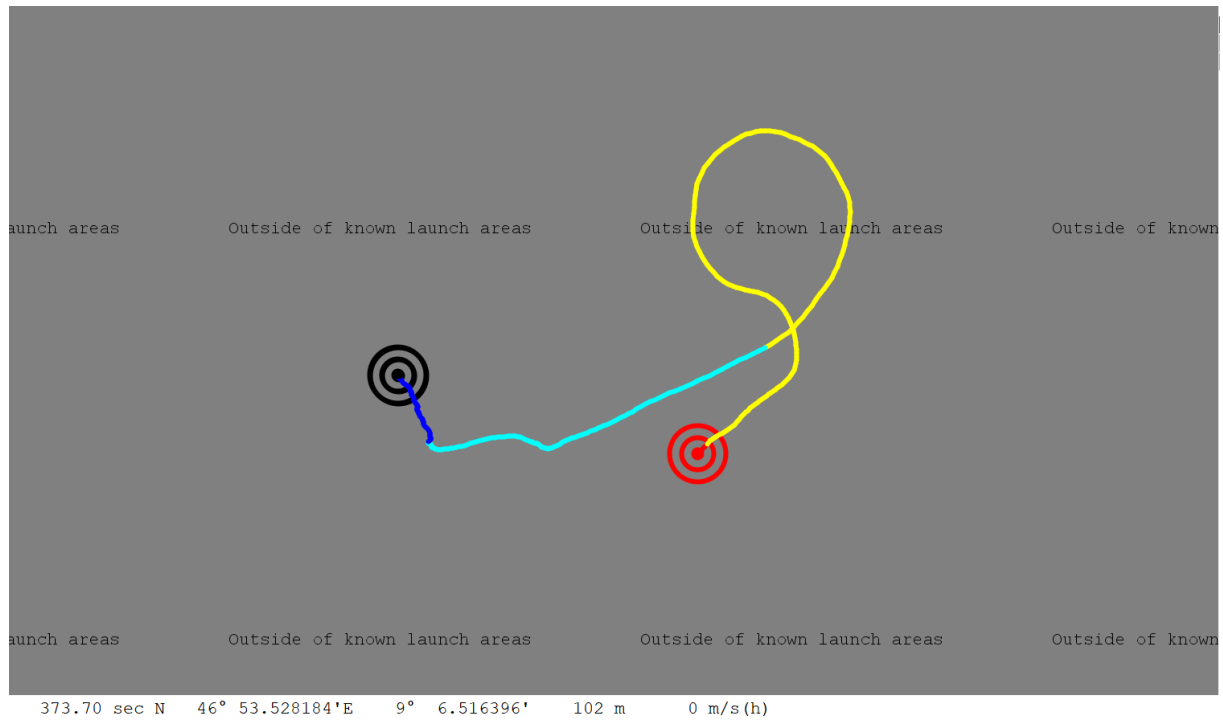


Figure 2.16 HDT 2: GPS Data obtained from TeleMega (COTS) FC

2.3.3 Final Ejection Tests

Purpose of Test:

Ejection tests are primarily performed to test the CO₂ separation mechanic. However, they also test the reliability of the electronics system, as the valves are actuated for each ejection which puts the system under stress. Actuating the valves requires a current of approx. 5A at 12V. Therefore, ejection tests double as load and stress tests for the electronics as well. These are the final ejection tests that were conducted.

Results:

Configuration	Consecutive Successes
Single Valve – TeleMega	10
Single Valve – CATS	10
Both Systems	5

Conclusion:

These tests were performed to assess the reliability of the CO₂ ejection mechanism. Nevertheless, it was also an opportunity to actuate the valves under pressure check the reliability in that scenario. With 15 consecutive successes on each side, the system may be called reliable.

3 Continuation

3.1 Reusability of Parts and Material

Item	Reusable/not Reusable/empty/etc
TeleMega 4.0 (1)	Pyro Channel C MosFET shorted, internal charging circuit non-functional REUSABLE
TeleMega 4.0 (2)	Pyro Channel A MosFET shorted REUSABLE
TeleMega 5.0	Pyro Channel C shorted REUSABLE
CATS (1)	Power Circuit MosFET fried (repaired by creator at time of writing the report), Pyro Channel 1 connector latch broke off during Argos Launch REUSABLE
CATS (2)	USB-C connector broke off (repaired by creator at time of writing the report) REUSABLE

3.2 How to Continue

The TeleMega flight computers have one malfunctional pyro channel each out of four pyro channels in total. Thus, they can be considered functional still, as the other 3 pyro channels have been tested and still function reliably. The damage to the pyros channels is strictly related to a failure mode which has been dealt with already.

As for the CATS flight computer, there are 3 pyro channels and 2 power circuits per board. Both CATS flight computers, as well as the three TeleMegas have been tested separately and independently for the same amount to ensure all of them are fully tested.

Apart of these, all other components are completely reusable and have not suffered any damage during any tests and launches/drops.

From this point on, reliability and functionality have been sufficiently tested and validated. Now, the work that is left is to change design for easier assembly and to conduct some additional reliability tests with said modified system. This is to ensure that nothing was compromised for easier accessibility and less confusing assembly steps for successful rocket system assembly.